

Saber Ayatollahi, CFA, MSc

Toronto, Canada • +1 (416) 856-1276 • saber.ayatollahi@gmail.com • www.linkedin.com/in/sayatollahi

SUMMARY

Senior Manager, ALM Risk Modelling candidate: ~7 years across asset-liability management and interest rate risk modelling for institutional portfolios, including delegated sign-off on client ALM outputs at Moody's Analytics, ALM/LDI studies at Ortec Finance, and financial-risk transformation delivery at EY. Led design of a multi-asset cash flow projection engine embedding behavioural and prepayment assumptions; built stochastic scenario generators; holds delegated sign-off on valuation, sensitivity, and ALM outputs for portfolios of \$5-25bn per engagement. Re-engineers legacy workflows into Python analytics pipelines with software-engineering discipline. CFA, dual MSc (Financial Modelling, Chemical Engineering).

CORE SKILLS

Structural Interest Rate Risk Modelling • Asset-Liability Management (ALM) • Behavioural Cash-Flow & Prepayment Modelling • Statistical & Stochastic Model Development • Yield Curve Construction & Spread Calibration • Derivatives Pricing Validation (Rates, FX, Inflation) • Python (pandas, NumPy, SciPy) & SQL • Software Engineering Practices (Git, CI/CD, Code Review) • Senior Stakeholder & Investment Committee Presentation

EXPERIENCE

Moody's Analytics

Asst. Director - Modelling Services (Investment & ALM Risk Lead)

Toronto, May 2023 - Present

Balance Sheet Interest Rate Risk Modelling

- Oversees interest rate risk and duration analysis for multi-asset institutional client portfolios under parallel and non-parallel rate shocks, aligned with IRRBB standards analogous to OSFI B-12 and Basel Committee frameworks.
- Holds delegated sign-off authority on valuation, sensitivity, and ALM outputs for institutional portfolios of \$5-25bn per engagement (~\$50bn+ cumulative), independently reviewing curve construction and spread calibration before production release.
- Validates derivatives pricing outputs across rates, FX, and inflation, confirming sensitivity and scenario consistency at portfolio-level ALM aggregates.

Behavioural Cash Flow Modelling & Balance Sheet Projection

- Led design and implementation of an enterprise multi-asset cash flow projection engine supporting base, stress, and reverse-stress scenarios, providing forward-looking liquidity visibility across ALM horizons.
- Embedded behavioural cash flow assumptions, prepayment logic, and macro stress overlays into the projection models, aligned with regulatory liquidity expectations.
- Architected configurable time-bucketed liquidity gap analytics (T+1 through multi-year) supporting funding optimization, refinancing risk assessment, and asset-allocation decisions.
- Reviews aggregation logic converting security-level exposures into portfolio-level risk metrics feeding downstream ALM and capital processes.

Model Coding, Software Practices & Stakeholder Engagement

- Re-engineered manual spreadsheet workflows into scalable Python analytics pipelines with embedded logging, validation, and auditability controls, maintained under Git version control.
- Built agentic AI development workflows (Claude Code, Cursor IDE) automating code generation, validation scaffolding, and anomaly detection, reducing development cycle time an estimated 30-40% on comparable modules.
- Prepares and presents analytical summaries on interest rate exposure, scenario impacts, and balance sheet sensitivities for senior stakeholders, translating quantitative outputs into decision-ready narratives.
- Serves on the model governance committee, contributing to methodology review, benchmarking, and documentation standards; escalates outputs lacking economic defensibility to clients' Heads of Risk.

Client Service Specialist

Toronto, May 2022 - May 2023

- Led onboarding and implementation for U.S. and Canadian pension funds and asset managers on PFaroe DB (risk & ALM analytics) and PFaroe PM, advising on asset, liability, and ALM configurations.
- Translated investment-team requirements into scoped development requests and validated model outputs post-deployment; migrated all assigned accounts off the legacy Calypso environment.

Ernst & Young (EY)

Senior Consultant - IFRS & Financial Risk Transformation

Toronto, Sep 2021 - Apr 2022

- Delivered IFRS 17 and IFRS 9 transformation projects for Canadian and international insurers, coordinating actuarial, finance, IT, and risk functions on data sourcing and model requirements.

Ortec Finance Canada

Consultant - Asset-Liability Management & Risk

Toronto, Feb 2019 - Sep 2021

- Built and interpreted stochastic economic scenario generators, calibrated to client assumptions, to quantify interest rate risk and funding volatility under base-case and stressed environments.
- Delivered ALM studies for Canadian and international pension funds and institutional investors, analyzing funding ratios, duration mismatch, and long-term balance sheet risk exposures.
- Modelled actuarial liabilities alongside all major investment product types and economic variables; performed cash flow, liquidity, and hedging analysis across interest rate, inflation, and currency risk.
- Developed and validated, on a team of three, the model capturing all Funding Policy dynamics for a three-plan university pension merger, evaluating duration, currency, inflation, and leverage overlay strategies.
- Presented model findings and allocation proposals to pension-fund investment committees on-site; ran asset-only and surplus optimization on VaR and CVaR with risk decomposition using Ortec's GLASS platform.

EDUCATION & CREDENTIALS

- Chartered Financial Analyst (CFA) - CFA Institute, 2024
- MSc, Financial Modelling - Western University, 2018
- MSc, Chemical Engineering (Full Scholarship) - Western University, 2016